09

Perimeter, area, surface area and volume

Part IV · Shape, structure and measure

CHAPTER 9 · OPENER

Perimeter, area, surface area and volume

Choosing the right measure—length, area, surface area or volume—connects geometric structure to material use, capacity and cost.

ENGAGEMENT

How can a community project estimate turf, paving, fencing, soil and paint without confusing boundary, surface and space?

Learning goals

- Distinguish perimeter, area, surface area and volume.
- Calculate with triangles, quadrilaterals, circles and composites.
- Use nets and prism formulas with correct units.
- Convert capacity, compare packaging and evaluate a project budget.

MYP lens

Key concept: Form

Related concepts: Measurement, Space

Global context: Globalization and sustainability

ATL focus. In order for a student to apply mathematics in real-life contexts, students must select the correct measure, convert units, calculate costs and evaluate a budget with stated assumptions.

Language bank: perimeter · area · surface area · volume · capacity · radius · diameter · circumference · semicircle · composite · net · prism · cross-section · litre

CHAPTER 9 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1	Find 7×4 and 2(7+4).	READY
R2	Find half of 8×5.	READY
R3	Calculate 6^2.	READY
R4	Convert 2 L to mL.	READY
R5	Find 1.2×0.6.	READY

Chapter map

1. **9.1 Perimeter and area of triangles and quadrilaterals** — Choose the quantity the task needs: boundary length, covered surface or space inside.
2. **9.2 Circles and composite shapes** — Identify radius and diameter, then split a composite shape into familiar pieces.
3. **9.3 Nets and surface area** — A net shows every outside face; surface area adds face areas, not the space inside.
4. **9.4 Prisms, volume and capacity** — Volume measures space; capacity measures how much a container can hold.

> **EXPLAIN HOW YOU KNOW**
>
> For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 9 · MATHEMATICAL DEVELOPMENT

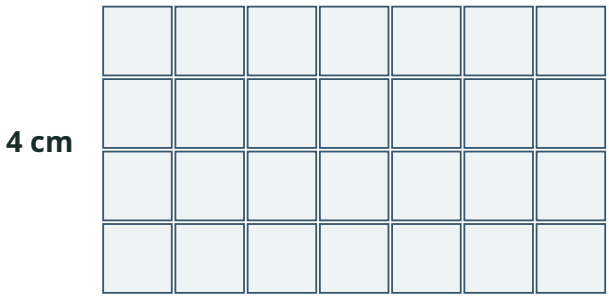
9.1 Perimeter and area of triangles and quadrilaterals

BIG IDEA

Choose the quantity the task needs: boundary length, covered surface or space inside.

Perimeter is the total boundary length. Area measures surface in square units. Rectangle area = length×width; triangle area = ½×base×perpendicular height; parallelogram area = base×height.

A trapezoid with parallel sides a and b and perpendicular height h has area ½(a+b)h. The height must be perpendicular to the parallel sides, not a slant edge.



Area = 28 cm² · Perimeter = 22 cm

Area counts square units (28 cm²); perimeter counts boundary units (22 cm).

WORKED EXAMPLE · A TRIANGLE AND A TRAPEZOID

Triangle: base 8 cm, height 5 cm. Trapezoid: parallel sides 6 and 10 cm, height 4 cm.

1. Triangle: ½×8×5 = 20 cm².
2. Trapezoid: ½×(6+10)×4 = 32 cm².

20 cm² and 32 cm².

CHAPTER 9 · PRACTICE

9.1 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Do not use a slant side as the height. Do not report area in cm or perimeter in cm^2.

GUIDED PRACTICE · FIRST STEPS

9.1.1 Find the area and perimeter of a 7 cm by 4 cm rectangle. FLUENCY

9.1.2 Find the area of a triangle with base 8 cm and perpendicular height 5 cm. FLUENCY

9.1.3 Find the area of a trapezoid with parallel sides 6 cm and 10 cm and height 4 cm. FLUENCY

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.1.4 A triangle has sides 5, 5, 6 cm. A student uses ½×6×5. Explain the error and find the correct height and area. REASONING

9.1.5 A rectangular lawn is 18 m by 12 m. Find its area and the cost of turf at \$7.50/m^2. APPLICATION

9.1.6 A rectangle has area 48 cm^2 and whole-number sides. Which dimensions give the smallest perimeter? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 9 · MATHEMATICAL DEVELOPMENT

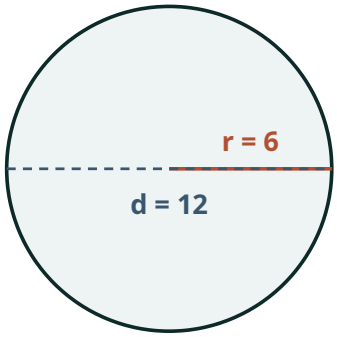
9.2 Circles and composite shapes

BIG IDEA

Identify radius and diameter, then split a composite shape into familiar pieces.

For a circle, diameter $d=2r$. Circumference $C=\pi d=2\pi r$; area $A=\pi r^2$. In this chapter use $\pi\approx3.14$ unless told otherwise, and label results as approximate.

A semicircle has half the circle area and a curved boundary of πr. For a composite plan, add areas of non-overlapping pieces and trace only the outer boundary for fencing.



$C = 2\pi r = \pi d \cdot A = \pi r^2 \cdot \pi \approx 3.14$

d = 2r. Area uses r^2; circumference uses 2r. Use $\pi \approx 3.14$ here.

WORKED EXAMPLE · A SEMICIRCULAR STAGE

Diameter 12 m, so radius 6 m.

1. Area = $\frac{1}{2}\times3.14\times6^2 = 56.52$ m^2.
2. Curved boundary = $3.14\times6 = 18.84$ m.

Area ≈56.52 m^2; arc ≈18.84 m.

CHAPTER 9 · PRACTICE

9.2 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

The diameter is not part of the outer boundary when the semicircle is attached to a rectangle along that edge.

GUIDED PRACTICE · FIRST STEPS

9.2.1 Find the circumference and area of a circle with radius 5 cm, using $\pi=3.14$. FLUENCY

9.2.2 A circle has diameter 12 m. Find its radius, circumference and area ($\pi=3.14$). FLUENCY

9.2.3 Find the area and curved boundary of a semicircle of diameter 12 m ($\pi=3.14$). APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.2.4 An 18 m by 12 m rectangle has a semicircle of diameter 12 m attached to one short end. Find total area and outer boundary ($\pi=3.14$). APPLICATION

9.2.5 Why is the 12 m diameter not included in the fencing length of the composite plan? REASONING

9.2.6 If the radius of a circle doubles, what happens to its circumference and area? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 9 · MATHEMATICAL DEVELOPMENT

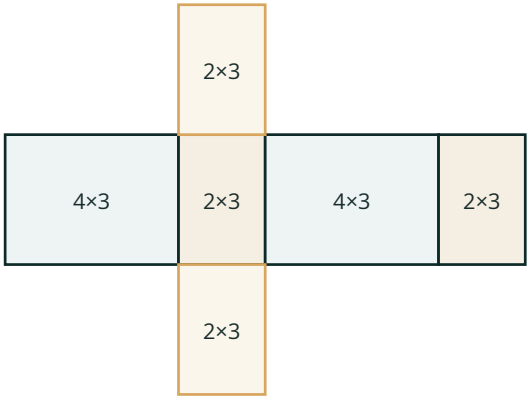
9.3 Nets and surface area

BIG IDEA

A net shows every outside face; surface area adds face areas, not the space inside.

A net is a flat pattern that folds into a solid. The surface area of a closed cuboid is 2(lw+lh+wh). For an open-topped box, omit the top face.

Use square units for surface area. Painting an outside surface depends on area; filling the box depends on volume. Keep the two quantities separate.



Surface area = 2(4×3 + 4×2 + 3×2) = 52 square units

A net exposes all six faces. Surface area = 2(lw + lh + wh).

WORKED EXAMPLE · AN OPEN-TOPPED RAISED BED

Length 1.2 m, width 0.6 m, height 0.5 m.

1. Base: $1.2 \times 0.6 = 0.72$ m^2.
2. Two long sides: $2 \times 1.2 \times 0.5 = 1.20$ m^2. Two short sides: $2 \times 0.6 \times 0.5 = 0.60$ m^2.

Outside area = 2.52 m^2.

CHAPTER 9 · PRACTICE

9.3 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A closed cuboid has six faces; an open-topped bed has five. Do not paint an imaginary top.

GUIDED PRACTICE · FIRST STEPS

9.3.1 Find the surface area of a closed cuboid 4 cm by 3 cm by 2 cm. FLUENCY

9.3.2 Find the surface area of a cube with side 6 cm. FLUENCY

9.3.3 Find the outside area of an open-topped bed 1.2 m by 0.6 m by 0.5 m. APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.3.4 A student uses the closed-cuboid formula for an open box. What must be subtracted? REASONING

9.3.5 Three beds each have outside area 2.52 m^2. Paint covers 9 m^2 per litre. How many litres are needed, and how many 1 L tins must be bought? APPLICATION

9.3.6 A cuboid has volume 24 cm^3 and whole-number edges. Compare surface areas of 1×1×24 and 2×3×4. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 9 · MATHEMATICAL DEVELOPMENT

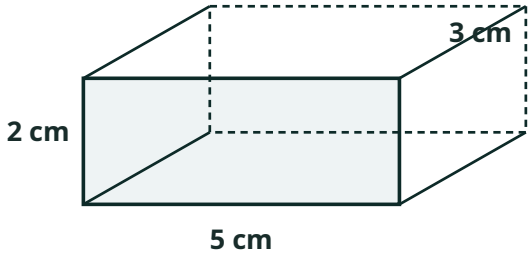
9.4 Prisms, volume and capacity

BIG IDEA

Volume measures space; capacity measures how much a container can hold.

A prism has a constant cross-section. Volume = cross-section area × length. For a cuboid, V=lwh; for a triangular prism, V=(½×base×height)×length.

Capacity conversions: 1 cm^3 = 1 mL; 1,000 cm^3 = 1 L; 1 m^3 = 1,000 L. Convert units before multiplying or dividing.



Volume = 5 × 3 × 2 = 30 cm^3 = 30 mL

Volume counts cubic units: 5 × 3 × 2 = 30 cm^3 = 30 mL.

WORKED EXAMPLE · A TANK AND A PRISM

Tank 50 cm × 30 cm × 40 cm; triangular prism with cross-section base 8 cm, height 5 cm, length 14 cm.

1. Tank volume 60,000 cm^3 = 60 L.
2. Prism volume ½×8×5×14 = 280 cm^3.

60 L; 280 cm^3.

CHAPTER 9 · PRACTICE

9.4 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

1 m^3 is 1,000 L, not 100 L. Length units scale by 100, area by 10,000 and volume by 1,000,000 between metres and centimetres.

GUIDED PRACTICE · FIRST STEPS

9.4.1 Find the volume of a cuboid 20 cm by 10 cm by 5 cm. FLUENCY

9.4.2 Find the volume of a triangular prism: cross-section base 8 cm, height 5 cm, length 14 cm. FLUENCY

9.4.3 A tank is 50 cm by 30 cm by 40 cm. Find its capacity in litres. APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

9.4.4 Convert 2.5 L to cm^3, 450 mL to cm^3 and 0.8 m^3 to litres. FLUENCY

9.4.5 A student says 1 m^3=100 L because 1 m=100 cm. Correct the reasoning. REASONING

9.4.6 A prism has volume 360 cm^3 and cross-section area 30 cm^2. Find its length. If every dimension doubles, what happens to volume? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 9 · EXPLORATION

The packaging problem

A box has volume 24 cm^3 and whole-number edges. Rotations count as the same design. Surface area measures cardboard needed for a closed box.

> **INVESTIGATION**
> Criterion B. Keep volume fixed, compare surface areas, test a second volume and justify a recommendation.

EDGES (CM)	VOLUME (CM^3)	SURFACE AREA (CM^2)
1×1×24	24	
1×2×12	24	
1×3×8	24	
1×4×6	24	
2×2×6	24	
2×3×4	24	

EXPLORATION · THE PACKAGING PROBLEM

Criterion B. Keep volume fixed, compare surface areas, test a second volume and justify a recommendation.

B1 Complete the surface areas for the six designs. INVESTIGATION

B2 Which design uses least and most cardboard? INVESTIGATION

B3 Test volume 27 cm^3 with 3×3×3, 1×3×9 and 1×1×27. Does the cube use least? INVESTIGATION

B4 Design two closed boxes of volume 36 cm^3 and recommend one by material use. CHALLENGE

B5 Give two practical reasons a manufacturer might not choose the minimum-area box, and one limitation of the pattern. INVESTIGATION

CHAPTER 9 · CHECKPOINT

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

C1	Find the area of a triangle with base 10 cm and height 6 cm.	CHECK
C2	Find the area of a circle with radius 3 cm (π=3.14).	CHECK
C3	Find the surface area of a cube with side 4 cm.	CHECK
C4	Find the capacity in litres of a 10 cm cube.	CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

CHAPTER 9 · APPLICATION

The courtyard project

An 18 m by 12 m rectangular lawn has a semicircular stage of diameter 12 m attached to one short end. Three open-topped beds are 1.2 m × 0.6 m × 0.5 m, filled to 80%. Budget $6,000.

REAL-LIFE APPLICATION

Criterion D. Simulated project. Use π=3.14, show units and round purchases up to whole bags or tins.

ITEM	PRICE	COVERAGE OR SIZE
Turf	$7.50/m^2	Lawn only
Paving	$42/m^2	Stage only
Fencing	$28/m	Outer boundary
Soil	$9.50 per 40 L bag	Each bed 80% full
Paint	$22 per 1 L tin	1 L covers 9 m^2

APPLICATION · THE COURTYARD PROJECT

Criterion D. Simulated project. Use π=3.14, show units and round purchases up to whole bags or tins.

D1 Find lawn area, stage area and total area. APPLICATION

D2 Find fencing length and the costs of turf, paving and fencing. APPLICATION

D3 Find soil needed for three beds and the number and cost of 40 L bags. APPLICATION

D4 Find paint needed for the three open beds and the tin cost. APPLICATION

D5 Find the total cost and budget position. Then add 10% contingency and state the new position. CHALLENGE

CHAPTER 9 · ASSESSMENT PRACTICE

The measure explainer

MYP ASSESSMENT PRACTICE

Criteria A and C. Explain which measure a purchase needs and check units with a small example.

A learner confuses perimeter, area, surface area and volume. Use a 4 cm by 3 cm by 2 cm box to make the distinctions clear.

ASSESSMENT PRACTICE · THE MEASURE EXPLAINER

Criteria A and C. Explain which measure a purchase needs and check units with a small example.

S1	Find the box volume and closed surface area, with units.	ASSESSMENT
S2	Which measure determines cardboard, paint and filling?	ASSESSMENT
S3	Write a unit-check warning and one conversion fact.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

CHAPTER 9 · MIXED PRACTICE

Mixed practice 9

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 9

Connect ideas. Give units and reasons where requested.

M1	Find the area of a parallelogram with base 9 cm and height 4 cm.	FLUENCY
M2	Find the circumference of a circle with diameter 10 cm (π=3.14).	FLUENCY
M3	Find the surface area of a closed cuboid 5 cm by 2 cm by 3 cm.	FLUENCY
M4	Find the capacity in litres of a tank 40 cm by 25 cm by 20 cm.	APPLICATION
M5	A student uses a slant side as triangle height. Explain the correction.	REASONING
M6	A semicircle has radius 4 m. Find its area and curved boundary (π=3.14).	APPLICATION
M7	A cuboid has volume 60 cm^3 and base area 12 cm^2. Find its height.	CHALLENGE
M8	If all lengths of a solid double, what happens to surface area and volume?	REASONING
M9	Paint covers 9 m^2/L. How many 1 L tins for 20 m^2?	APPLICATION
M10	Compare closed boxes 1×1×36 and 3×3×4 (both volume 36). Which uses less cardboard and by how much?	CHALLENGE

FINAL REFLECTION

Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 9 · ANSWERS AND REASONING

Answers · Ready check and Checkpoint

READY CHECK

R1 **28 and 22.**
Area and perimeter of a 7 by 4 rectangle.

R2 **20.**
½×40.

R3 **36.**
6×6.

R4 **2,000 mL.**
1 L=1,000 mL.

R5 **0.72.**
Multiply as whole numbers, then place two decimal digits.

CHECKPOINT

C1 **30 cm^2.**
½×10×6.

C2 **28.26 cm^2.**
3.14×9.

C3 **96 cm^2.**
6×16.

C4 **1 L.**
1,000 cm^3.

CHAPTER 9 · ANSWERS AND REASONING

Answers · 9.1 and 9.2

PRACTICE 9.1 · AREA AND PERIMETER

9.1.1 **28 cm²; 22 cm.**
Area 7×4; perimeter 2(7+4).

9.1.2 **20 cm².**
½×8×5.

9.1.3 **32 cm².**
½(6+10)×4.

9.1.4 **The 5 cm side is slant, not perpendicular height. Height is 4 cm; area 12 cm².**
Half the base is 3; 3–4–5 right triangle.

9.1.5 **216 m²; $1,620.**
18×12; multiply by 7.50.

9.1.6 **6 cm by 8 cm; perimeter 28 cm.**
Compare factor pairs of 48; the closest pair minimises perimeter.

PRACTICE 9.2 · CIRCLES AND COMPOSITES

9.2.1 **31.4 cm; 78.5 cm².**
$2\pi r$ and πr^2.

9.2.2 **6 m; 37.68 m; 113.04 m².**
$r=d/2$; $C=\pi d$; $A=\pi r^2$.

9.2.3 **56.52 m²; 18.84 m.**
Half the circle area; half the circumference.

9.2.4 **272.52 m²; 66.84 m.**
Area 216+56.52; boundary 18+18+12+18.84.

9.2.5 **It is an internal join between lawn and stage, not an outer edge.**
Fencing follows the outside boundary only.

9.2.6 **Circumference doubles; area becomes four times as large.**
C is proportional to r; A to r^2.

CHAPTER 9 · ANSWERS AND REASONING

Answers · 9.3 and 9.4

PRACTICE 9.3 · SURFACE AREA

9.3.1 **52 cm^2.**
$2(12+8+6)$.

9.3.2 **216 cm^2.**
6×6^2.

9.3.3 **2.52 m^2.**
Base 0.72 + long sides 1.20 + short sides 0.60.

9.3.4 **The top area lw.**
The top is not an outside face to paint.

9.3.5 **0.84 L needed; buy 1 tin.**
$3\times2.52=7.56$ m^2; $7.56\div9=0.84$; round tins up.

9.3.6 **98 cm^2 and 52 cm^2. The compact box uses less material.**
Same volume does not imply same surface area.

PRACTICE 9.4 · VOLUME AND CAPACITY

9.4.1 **1,000 cm^3 = 1 L.**
$20\times10\times5$.

9.4.2 **280 cm^3.**
$\frac{1}{2}\times8\times5\times14$.

9.4.3 **60 L.**
60,000 $cm^3\div1{,}000$.

9.4.4 **2,500 cm^3; 450 cm^3; 800 L.**
Use 1 L=1,000 cm^3 and 1 m^3=1,000 L.

9.4.5 **1 m^3=1,000,000 cm^3=1,000 L. Volume scales in three dimensions.**
Cube the length conversion.

9.4.6 **12 cm; volume becomes 8 times as large.**
Length=V÷A; scale factor $2^3=8$.

CHAPTER 9 · ANSWERS AND REASONING

Answers · Exploration

EXPLORATION · THE PACKAGING PROBLEM

B1 **98, 76, 70, 68, 56, 52 cm^2.**
Use 2(lw+lh+wh).

B2 **Least 2×3×4 (52 cm^2); most 1×1×24 (98 cm^2).**
Compare the completed surface areas.

B3 **Surface areas 54, 78, 110 cm^2; yes, the cube uses least.**
Compact dimensions reduce exposed face area.

B4 **Sample: 3×3×4 has area 66 cm^2; 1×6×6 has area 96 cm^2. Recommend 3×3×4 for less cardboard.**
Both volumes are 36; compare surface areas.

B5 **Sample: product shape, stacking, opening strength or shelf display may matter. The pattern compares closed boxes with whole-number edges and ignores waste, tabs and thickness.**
A mathematical optimum is not the only design criterion.

CHAPTER 9 · ANSWERS AND REASONING

Answers · Application and Assessment

APPLICATION · THE COURTYARD PROJECT

D1 **216 m^2; 56.52 m^2; 272.52 m^2.**
Rectangle plus semicircle of radius 6 m.

D2 **66.84 m; turf $1,620; paving $2,373.84; fencing $1,871.52.**
Boundary 18+18+12+18.84; multiply each quantity by its price.

D3 **864 L; 22 bags; $209.**
One bed 0.36 m^3=360 L; 80% is 288 L; three beds 864 L; ceil(864/40)=22.

D4 **7.56 m^2 outside area; 0.84 L paint; buy 1 tin for $22.**
One bed 2.52 m^2; three beds 7.56; divide by 9 and round tins up.

D5 **$6,096.36, over by $96.36. With 10% contingency: $6,706.00 (to the nearest cent), over by $706.00.**
Add 1620+2373.84+1871.52+209+22; multiply by 1.10.

ASSESSMENT PRACTICE · THE MEASURE EXPLAINER

S1 **24 cm^3; 52 cm^2.**
4×3×2; 2(12+8+6).

S2 **Cardboard and paint use surface area; filling uses volume or capacity.**
Boundary length would use perimeter.

S3 **Sample: "Area uses square units, volume cubic units; 1,000 cm^3 = 1 L."**
Units reveal whether a formula matches the quantity.

CHAPTER 9 · ANSWERS AND REASONING

Answers · Mixed practice

MIXED PRACTICE 9

M1 **36 cm².**
base×height.

M2 **31.4 cm.**
πd.

M3 **62 cm².**
2(10+15+6).

M4 **20 L.**
20,000 cm³÷1,000.

M5 **Use the perpendicular distance from the base to the opposite vertex; a slant side is not the height unless it is perpendicular.**
Area requires base×perpendicular height÷2.

M6 **25.12 m²; 12.56 m.**
½πr² and πr.

M7 **5 cm.**
60÷12.

M8 **Surface area ×4; volume ×8.**
Area scales with the square, volume with the cube.

M9 **3 tins.**
20÷9≈2.22; round up.

M10 **3×3×4: 66 cm² versus 146 cm², saving 80 cm².**
Use 2(lw+lh+wh).